

Quantitative trait locus analysis for days-to-heading and morphological traits in an RIL population derived from an extremely late flowering F₁ hybrid of sorghum

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Abstract The hybrid vigor typical of F₁ cultivars is used to boost biomass production of sorghum (*Sorghum bicolor* (L.) Moench). The high dry-matter yielding F₁ cultivar Kazetachi uniquely shows extremely late flowering and a long culm, and is greatly different from its parents. We investigated the genetic mechanisms underlying these phenotypes by quantitative trait locus

(QTL) analysis of recombinant inbred lines derived from a male-fertile line and a restorer line and grown in 3 years. QTL analysis for six traits (days-to-heading, culm length, culm width, culm number, panicle length, panicle number) revealed that the unique phenotypes of the F₁ plants were controlled by the genetic combination of 12 or more QTLs detected in at least 2 years. Two putative QTLs for days-to-heading (*qDH1* on SBI-01 and *qDH6* on SBI-06) would strongly affect the other phenotypes because of their co-localization with QTLs for other traits, as supported by significant phenotypic correlations. These QTLs would be useful for understanding the association of plant type with biomass production in sorghum.

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Introduction

Sorghum (*Sorghum bicolor* (L.) Moench) originated in northeast Africa and is grown worldwide for food and forage between the latitudes of 40°S and 45°N, primarily in drier regions. Diversification has allowed sorghum to adapt to various environmental conditions. The core collection of the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) shows wide diversity in heading date, plant height, and tiller number (Upadhyaya et al. 2009). The genetic combination of this diversity could lead to sustainable high

yields for several uses, including grain for food, foliage for forage, building materials, and juice for sweet syrup (Smith and Frederiksen 2000).

Population growth and changes in dietary habits have increased the demand for cereal production for food and forage. Additionally, competition by biomass production for bioethanol manufacture (Farrell et al. 2006) has increased the price of and demand for cereals. As sorghum needs less water than maize, it could be grown in semi-arid areas unfit for the cultivation of maize. To this end, genetic resources for breeding have been evaluated (Almodares and Hadi 2009; Srinivasa Rao et al. 2009; Wang et al. 2009).

Sorghum is typically bred through the use of a male-sterile line and a restorer line to create F_1 hybrids. Combinations that show hybrid vigor are used (Smith and Frederiksen 2000). Some combinations show hybrid vigor in plant height or in maturity relative to the mid-parent date in the range of -8 to $+20$ days (Bartel 1949). Recessive homozygous alleles at four loci (*dw1-dw4*) cause dwarfing (Quinby and Karper 1954). The dominant alleles could therefore result in longer culms and greater foliage yield. An F_1 line that does not flower under natural long-day conditions in middle latitudes (Rooney and Aydin 1999) should produce high foliage yields because of its termless vegetative growth habit, and paves the way for the use of genetically modified plants that cannot release pollen or seed.

Molecular studies of sorghum have detected several quantitative trait locus (QTL) for agronomic traits related to biomass production, and have identified a few genes, including the maturation genes *Ma3* (Childs et al. 1997) and *Mal* (Murphy et al. 2011) and the dwarfing gene *dw3* (Multani et al. 2003). However, the identification and function of some genes associated with maturity and culm length (Quinby and Karper 1954; Quinby 1966) remain unclear and must be elucidated at the molecular level (Mace and Jordan 2010).

In Japan, the F_1 cultivar Kazetachi, which is derived from early-flowering parents, shows extremely late flowering under long-day condition and hybrid vigor in plant size, and is used for forage production (Tarumoto et al. 2000). Our aims were to clarify the underlying genetic mechanisms by using recombinant inbred lines (RILs) derived from the parents, to compare the QTLs of biomass-related traits with

QTLs found in previous studies, and to explore the genetic interactions that could increase total biomass production.

Materials and methods

Plant materials

The experimental materials consisted of a set of 136 $F_{8:10}$ RILs derived from the cross of MS138B with 74LH3213 (Supp. Fig. S1 in Online Resource 1). A set of RILs has been grown and harvested for seed in the field every year. Late-heading RILs were transplanted into pots in a greenhouse and harvested for seed in winter under natural short-day condition.

The female parent, MS138B, is a maintainer line (B-line) bred at Purdue University. To examine the agronomic traits of the RILs, we used MS138B in place of the male-sterile line MS138A as the RIL parent because segregation of sterility would bias allele frequencies in the RIL population. MS138A has been crossed many times with MS138B for seed production, and few differences between them in agronomic traits such as days-to-heading have been observed (data not shown). So the F_1 of MS138B $\times$ 74LH3213 is essentially no different from Kazetachi (MS138A $\times$ 74LH3213).

The male parent, 74LH3213, is a restorer line (R-line) bred at the International Center for the Improvement of Maize and Wheat (CIMMYT). Kazetachi (MS138A $\times$ 74LH3213) was first bred at Nagano Animal Industry Experiment Station in 1993. Kazetachi currently accounts for the largest proportion of sorghum seeds sold in Japan.

Field design and measurements

The parents, Kazetachi, and the RILs were planted at the National Agricultural Research Center for Kyushu Okinawa Region (KONARC; 130°44'E, 32°53'N) in 2008, 2009, and 2010. The mean day length during the growing period was 13.9 h in mid-May, 14.3 h in June, 14.1 h in July, 13.4 h in August, 12.4 h in September, and 11.4 h in October. The 24 hr average temperatures were 20.3 °C in May, 23.6 °C in June, 27.7 °C in July, 28.9 °C in August, 25.8 °C in September, and 19.8 °C in October.

The experimental design used one plot per line (no replication), consisting of 24 plants 0.08 m apart in a 2 m row 0.8 m from the next row (15.625 plants/m²). Five plants with intermediate growth were used for the phenotypic evaluation. Each plot received a basal dressing of 100–44–17 kg N–P–K, 1,000 kg magnesium–calcium, 1,000 kg fused magnesium phosphate, and 50,000 kg farmyard manure per ha before sowing. Days-to-heading, culm length, culm width, panicle length, culm number at base, and number of panicles were measured on each plant. Days-to-heading was recorded as days from sowing to heading. Culm length was measured from the soil surface to the neck of the panicle. Culm width was measured 5 cm above the soil surface. Panicle length was measured from the neck to the top of the panicle. Culms and panicles were counted 1 month after heading or at end of October.

Correlation analysis and QTL analysis used the middle three values from each group of five plants. Correlation analysis was performed in JMP v. 9.0 software (SAS Institute Inc., Cary, NC, USA). The expected-mean-squares method was used to fit phenotypic traits to a linear model, and Pearson's phenotypic correlation coefficients were calculated with the "density ellipse" command for each pair of traits among all six traits.

Genotyping of RILs

Genomic DNAs were extracted from seedlings of the 136 F₈ RILs. We selected 240 simple sequence repeat (SSR) primers showing polymorphisms between the parents from published SSR markers (Bhatramakki et al. 2000; Yonemaru et al. 2009). Additionally, five SSR markers, one cleaved amplified polymorphic sequence (CAPS) marker, and one allele-specific marker were newly developed. All were used for genotyping and constructing a linkage map (Supp. Table S1 in Online Resource 2).

SSRs were PCR-amplified in a 20 µL reaction mixture consisting of 20 ng DNA template, 20 ng of each primer, 10× PCR buffer (20 mM Mg²⁺), 2.5 mM dNTP mixture, and 1 unit of Taq DNA polymerase (Takara Co., Ltd., Otsu, Japan) in a Veriti 96-well thermal cycler (Applied Biosystems, Foster City, CA, USA). The conditions were an initial denaturation step of 3 min at 94 °C, then 32 cycles of 1 min at 94 °C, 1 min at 55 °C, and 45 s at 72 °C, with no final extension phase. For PCR using CAPS

and allele-specific markers, the conditions consisted of an initial 3 min at 94 °C; 32 cycles of 1 min at 94 °C, 1 min at 54 °C, and 2 or 1.5 min at 72 °C; and a final 7 min at 72 °C. The restriction endonuclease used for the CAPS marker was *Hind*III. Each PCR product, both with and without digestion, was analyzed by gel electrophoresis in 2 or 4 % agarose gel in TAE (Tris, acetic acid, EDTA) buffer. After staining with ethidium bromide, the banding patterns were photographed under ultraviolet light for genotyping.

Linkage and QTL mapping

Linkage mapping was performed with MAPMAKER/EXP v. 3.0 software (Lander et al. 1987). The Kosambi mapping function (Kosambi 1944) was used to convert recombination into genetic distance in centimorgans (cM). QTL analyses were performed using composite interval mapping as implemented by the program Zmapqtl (model 6) of the QTL Cartographer v. 2.5 software (Wang et al. 2011). Genome-wide threshold values ($\alpha = 0.05$) were used to detect putative QTLs based on the results of 1,000 permutations (Churchill and Doerge 1994). When separated by a minimum distance of 20 cM or a 2-LOD support interval from peak to valley, neighboring peaks were considered to be different QTLs. Otherwise, the higher peak was chosen as the position of the QTL. The confidence interval for each QTL was set at the 1-LOD support interval.

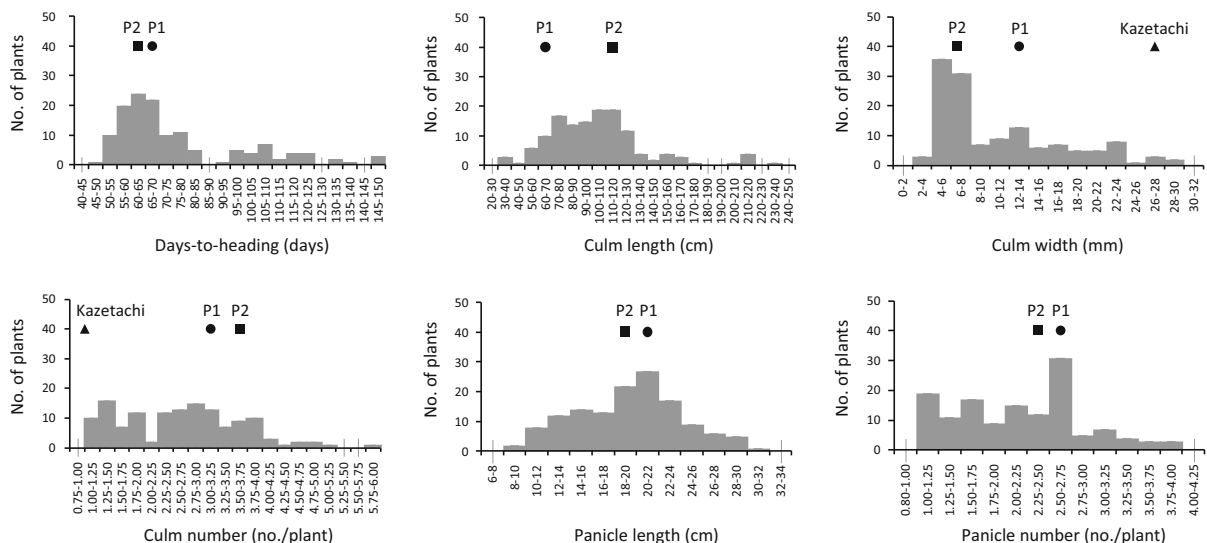
Results

Data analysis of six traits related to biomass

Kazetachi had not headed by 150 days in 2008 and 2009, so days-to-heading and panicle number and length could not be recorded. The mean values of culm length and width were $\geq 2\times$ the parental values (Table 1). On the other hand, Kazetachi had only one main culm, whereas the parental lines had three or more. All six traits showed transgressive segregation among the RILs, but few plants were similar in type to Kazetachi (Table 1; Fig. 1). Days-to-heading was similarly short in both parental lines; in contrast, the RILs showed a wide range, from very short to very long. The distribution of culm width was similarly wide, but more RILs had a narrower culm than had a

Table 1 Mean trait values for parental lines, Kazetachi, and recombinant inbred lines (RILs) in 3 years (2008–2010)

Trait	Parental lines ^a			RIL population			
	MS138B	74LH3213	Kazetachi ^b	Min.	Max.	Mean	SD ^d
Days-to-heading (DH)	67.5	62.5	– ^c	47.5	148.4	77.0	23.4
Culm length (CL) (cm)	64.0	114.6	370.0	38.2	237.2	103.3	36.9
Culm width (CW) (mm)	12.9	6.6	26.4	3.4	28.6	11.0	6.6
Culm no. (CN) (/plant)	3.1	3.6	1.0	1.1	5.9	2.6	1.0
Panicle length (PL) (cm)	20.7	19.1	–	9.0	31.6	19.4	4.8
Panicle no. (PN) (/plant)	2.4	3.1	1.0 ^e	1.1	4.0	2.2	0.7

^a Means were calculated from 2009 and 2010 data^b Means were calculated from 2008 and 2009 data^c No data recorded because of extremely late heading^d Standard deviation^e Number of young panicles waiting to emerge**Fig. 1** Frequency distributions of the mean values of days-to-heading and five morphological traits among 136 recombinant inbred lines derived from MS138B (P₁) × 74LH3213 (P₂) infield experiments over 3 years (2008–2010). The symbol for Kazetachi marks the mean value of the F₁ (MS138A × 74LH3213)

wider culm relative to the parents. The distributions of culm length and panicle length resembled a normal distribution.

All pairs of traits were significantly correlated ($P < 0.01$, Table 2). Days-to-heading had positive correlations with culm width, culm length, and panicle length, and negative correlations with both culm and panicle numbers. Culm and panicle numbers each had negative correlations with culm and panicle lengths.

Construction of linkage map and QTL identification

Each RIL plant was successfully genotyped. A genetic map was constructed on the basis of the 247 markers, and 10 linkage groups (LGs) corresponding to the 10 sorghum chromosomes (Kim et al. 2005) were mapped with no gaps (Fig. 2). The total map length was 1,697.2 cM, with an average interval between markers of 6.9 cM.

Table 2 Correlations among six biomass-related traits in the RIL population

Correlation	CL	CW	CN	PL	Panicle No.
Days-to-heading (DH)	0.25*	0.78*	−0.40*	0.30*	−0.39*
Culm length (CL)		0.22*	−0.18*	0.35*	−0.11*
Culm width (CW)			−0.36*	0.44*	−0.34*
Culm no. (CN)				−0.23*	0.55*
Panicle length (PL)					−0.25*

* $P < 0.01$

Composite interval mapping identified 21 putative QTLs significantly associated with biomass-related traits in seven LGs (except SBI-02, SBI-03, and SBI-05) in 3 years, of which 12 QTLs were detected in at least 2 years (Fig. 2; Table 3).

Three QTLs for days-to-heading were detected on LGs SBI-01, SBI-06, and SBI-10 in 2 or 3 years. *qDH1* (SBI-01, 2008–2010) had a peak LOD score of 22.7–26.1 and accounted for the highest percentage of phenotypic variance explained (PVE: 44.24–49.63 %). *qDH6* (SBI-6, 2008–2010) had the second-highest LOD score and delayed heading by the additive effect of 74LH3213. *qDH10* (SBI-10, 2008, 2010) had the lowest score.

Four QTLs for culm length were detected on LGs SBI-06, SBI-07, and SBI-09. *qCL6.1* (2009) and *qCL6.2* (2008) were located near *qDH6* with LOD scores of 7.1 and 7.4, respectively, and PVE values of 16.39 and 14.50 %. *qCL7* (2008–2010) accounted for the highest PVE (20.10–23.65 %). *qCL9* was found in all 3 years.

Two QTLs for culm width were detected in all 3 years. *qCW1* (SBI-01) had much higher LOD scores and PVE values in 2008 and 2010 than in 2009. It was co-localized with *qDH1* between markers SB658 and SB718. *qCW6* (SBI-06) was detected near *qDH6*.

Two QTLs for culm number were detected on LGs SBI-01 and SBI-06. *qCN1* (2008–2010) had higher values of PVE than *qCN6*. *qCN1* was mapped near *qDH1*. *qCN6* (2009) had the lowest LOD score of all the QTLs detected.

Four QTLs for panicle length (*qPL1.1*, *qPL1.2*, *qPL6*, and *qPL7*) and six for panicle number (*qPN1*, *qPN4*, *qPN6*, *qPN8.1*, *qPN8.2*, and *qPN10*) were identified. *qPL1.2*, *qPL6*, and *qPL7* (2008–2010) were co-localized with QTLs for other traits: *qPL1.2* with *qDH1*, *qCW1*, *qCN1*, and *qPN1*; *qPL6* with *qDH6*, *qCL6.2*, *qCW6*, and *qPN6*; and *qPL7* with *qCL7*,

shortening panicle and culm lengths by the additive effect of the 74LH3213 alleles. *qPN10* was co-localized with *qDH10*. *qPL1.1*, *qPN4*, *qPN8.1*, and *qPN8.2* were each found in 1 year.

Discussion

QTLs for days-to-heading

Plant maturation affects the period of growth and thus determines biomass at harvest if there is no marked change in growth after heading. The late maturation of Kazetachi is due to the combination of at least three putative QTLs, *qDH1*, *qDH6*, and *qDH10*. The magnitude and direction of the additive effects of these QTLs imply that late maturation is due to the complementary interaction of the dominant MS138B functional allele of *qDH1* and the dominant 74LH3213 functional alleles of *qDH6* and *qDH10*. Notably, *qDH1* and *qDH6* together accounted for >60 % of PVE and thus play a major role in late maturation.

Ma1–Ma4, a series of maturity-related photoperiod-sensitivity genes (Quinby 1967), cannot fully explain late maturation. Rooney and Aydin (1999) proposed the addition of *Ma5* and *Ma6*. Reduced day length achieved by delayed sowing advanced vegetative maturation toward the parental values (Tarumoto et al. 2000), so a photoperiod-sensitivity gene could delay the maturation of Kazetachi under natural long-day condition. Therefore, the late maturation of Kazetachi might be controlled by the combination of several photoperiod-sensitivity genes.

Ma3 encodes a phytochrome B (Childs et al. 1997). The physical position of *Ma3* is estimated at 60.9 Mb in LG SBI-01 from sorghum genome information (Paterson et al. 2009). This position is very close to

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Table 3 QTLs for six biomass-related traits detected in 3 years

Trait	Years	QTL name	Chr.	Marker interval	Nearest marker	LOD score	Additive effect ^a	% PVE ^b
Days-to-heading	2008	<i>qDH1</i>	1	SB658–SB718	SB672	22.7	16.21	44.24
		<i>qDH6</i>	6	SB23991–SB24009	SB23996	11.3	−10.29	17.47
		<i>qDH10</i>	10	SB5242–SB5277	SB5257	3.5	−5.41	4.78
	2009	<i>qDH1</i>	1	SB658–SB678	SB672	23.0	16.50	49.63
		<i>qDH6</i>	6	SB23991–SB3398	SB23996	10.4	−9.34	15.74
		<i>qDH1</i>	1	SB658–SB678	SB672	26.1	18.40	48.98
	2010	<i>qDH6</i>	6	SB3387–SB24009	SB23996	10.1	−9.93	14.08
		<i>qDH10</i>	10	SB5242–SB5277	SB5257	3.5	−5.60	4.39
		<i>qCL6.2</i>	6	SB23991–SB3420	R1854	7.4	−15.28	14.50
		<i>qCL7</i>	7	SB4077–SB4094	SB4090	8.1	−19.28	23.65
Culm length	2008	<i>qCL9</i>	9	SB5031–SB5088	SB5042	5.0	14.04	12.27
		<i>qCL6.1</i>	6	SB3377–SB3390	SB3385	7.1	−16.00	16.39
		<i>qCL7</i>	7	SB4077–SB4094	SB4090	7.9	−17.64	22.42
	2009	<i>qCL9</i>	9	SB5031–SB5088	SB5059	9.2	15.95	16.72
		<i>qCL7</i>	7	SB4077–SB4094	SB4090	6.3	−16.85	20.10
		<i>qCL9</i>	9	SB5042–SB5088	SB5053	9.7	17.41	19.97
		<i>qCL7</i>	7	SB4077–SB4094	SB4090	6.3	−16.85	20.10
	2010	<i>qCL9</i>	9	SB5042–SB5088	SB5053	9.7	17.41	19.97
		<i>qCW1</i>	1	SB658–SB678	SB672	27.7	5.86	56.73
		<i>qCW6</i>	6	SB23991–SB24009	SB23996	10.3	−2.58	11.08
Culm width	2008	<i>qCW1</i>	1	SB658–SB718	SB672	17.0	3.95	27.80
		<i>qCW6</i>	6	SB3395–SB3424	R1854	10.9	−2.39	14.98
		<i>qCW1</i>	1	SB658–SB678	SB672	27.3	5.60	58.52
	2009	<i>qCW6</i>	6	SB23991–SB3398	SB23996	10.8	−2.80	13.39
		<i>qCN1</i>	1	SB644–SB718	SB672	7.0	−0.43	16.06
		<i>qCN1</i>	1	SB658–SB718	SB672	7.9	−0.47	18.72
		<i>qCN6</i>	6	SB24931–SB3528	SB3498	3.0	−0.28	6.54
	2010	<i>qCN1</i>	1	SB636–SB672	SB650	9.7	−0.54	23.23
		<i>qPL1.1</i>	1	SB317–SB419	SB369	3.4	−1.37	7.34
		<i>qPL1.2</i>	1	SB644–SB718	SB678	11.1	2.58	22.22
Panicle length	2008	<i>qPL6</i>	6	SB23991–SB3409	SB23996	10.1	−2.38	19.86
		<i>qPL7</i>	7	SB4077–SB4124	SB4094	7.0	−1.79	12.97
		<i>qPL1.2</i>	1	SB658–SB718	SB678	9.4	2.36	20.27
		<i>qPL6</i>	6	SB23991–SB24009	SB23996	10.1	−2.46	21.93
	2009	<i>qPL7</i>	7	SB4077–SB4137	SB4094	4.8	−1.54	9.49
		<i>qPL1.2</i>	1	SB658–SB718	SB678	11.3	2.52	22.50
		<i>qPL6</i>	6	SB23991–SB3398	SB23996	10.7	−2.59	21.11
		<i>qPL7</i>	7	SB4077–SB4124	SB4094	5.2	−1.56	9.49
	2010	<i>qPN1</i>	1	SB644–SB718	SB672	6.6	−0.36	13.21
		<i>qPN4</i>	4	SB2871–SB2961	SB2938	5.7	−0.38	14.48
Panicle number	2008	<i>qPN6</i>	6	SB3387–SB3398	SB23996	7.6	0.42	15.55
		<i>qPN8.1</i>	8	SB4342–SB4417	SB4392	3.3	−0.17	7.36
		<i>qPN8.2</i>	8	SB4635–SB4652	SB4642	6.6	−0.25	15.87
	2009	<i>qPN10</i>	10	SB5242–SB5277	SB5268	3.5	0.18	8.37
		<i>qPN1</i>	1	SB658–SB718	SB672	9.2	−0.53	20.66
		<i>qPN1</i>	1	SB658–SB718	SB672	9.2	−0.53	20.66
		<i>qPN1</i>	1	SB658–SB718	SB672	9.2	−0.53	20.66
	2010	<i>qPN1</i>	1	SB658–SB718	SB672	9.2	−0.53	20.66
		<i>qPN1</i>	1	SB658–SB718	SB672	9.2	−0.53	20.66
		<i>qPN1</i>	1	SB658–SB718	SB672	9.2	−0.53	20.66

^a Positive values show that the MS138B allele increases values^b Percentage of phenotypic variance explained

but a fine-mapping study placed *Ma1* on the long arm of SBI-06, away from *qDH6* (Klein et al. 2008; Murphy et al. 2011). Therefore, we consider that *qDH6* is different from *Ma1*. Additionally, *Ma5* (Rooney and Aydin 1999) was positioned at 152.5–166.0 cM on SBI-02 (Mace and Jordan 2010). So *qDH6* might be *Ma2* or *Ma6*. Whether *qDH6* is novel or a known QTL, the dominant allele would have strong photoperiod sensitivity and contribute to the lateness of Kazetachi by complementary interaction with *qDH1* and *qDH10*.

QTLs for culm length

We detected four culm length QTLs, two on the short arm of SBI-06 and one each on the long arms of SBI-07 and SBI-09. Despite their different positions, *qCL6.1* and *qCL6.2* could be the same QTL because of their similar LOD scores and additive effects. The direction of the additive effects could indicate the function of alleles: elongation of culm length by the 74LH3213 alleles of *qCL6.1*, *qCL6.2*, and *qCL7* and by the MS138B allele of *qCL9*. The slight difference in the additive effects between the 74LH3213 and MS138B alleles could reflect the longer culm length of 74LH3213 (Table 1; Supp. Fig. S1 in Online Resource 1).

The co-localization of *qCL6.1*, *qCL6.2*, and *qDH6* at the top of SBI-06 might be different from that of *dw2*, which is mapped on the long arm of SBI-06 (Klein et al. 2008). *qCL6.1*, *qCL6.2*, and *qDH6* may represent a single locus at which the 74LH3213 allele increases both days-to-heading and culm length. The significant positive correlation between days-to-heading and culm length is consistent with this hypothesis. *qCL7* could be distinct from *dw3* (Brown et al. 2008), but fine mapping will be essential for accurate comparison. *qCL9*, the MS138B allele of which extends culm length, would be the same QTL as Sb.Ht9.1 reported by Brown et al. (2008).

Kazetachi, the F_1 of MS138A $\times$ 74LH3213, which was virtually identical to the F_1 of MS138B $\times$ 74LH3213, had a culm length four times that of the midway length between MS138B and 74LH3213 (Table 1). The hypothesis that heterozygous alleles of three QTLs (*qCL6.1(2)*, *qCL7*, and *qCL9*) controlling culm length gave a major effect on the culm length of the RIL population (Table 1). Thus, the F_1 hybrid of parental lines with dwarfing alleles at different loci would have a much longer culm than the parents and could be valuable for biomass production.

QTLs for panicle length

Panicle length is an important component of kernel weight per panicle (Rami et al. 1998). More than 10 QTLs for the inflorescence architecture of sorghum have been reported (Brown et al. 2006).

We detected four putative QTLs for panicle length. *qPL1.1* was detected only in 2008. *qPL1.2* and *qPL6* were co-localized, respectively, with *qDH1* and *qDH6*, for days-to-anthesis. The significant positive correlation between days-to-anthesis and panicle length implies that *qPL1.2* and *qPL6* are the same as *qDH1* and *qDH6*. On the other hand, *qPL7* was detected at the same position as *qCL7*, for culm length. Mace and Jordan (2011) reported high QTL density surrounding the location of the major height gene *dw3*, including QTLs for biomass yield, lodging, panicle architecture, sugar-related traits, and green bug resistance. Similarly, *qPL6* was co-localized with *qCL6.2*, for culm length. On the other hand, *qPL1.2* was not co-localized with QTLs for culm length. We suppose *qPL1.2* to be different from the QTLs for culm length. This result allows for the possibility of plant types with longer panicles and shorter culm length and vice versa.

QTLs for culm width

A wide culm may contribute to biomass production. We found two putative QTLs for culm width near *qDH1* and *qDH6*, for days-to-heading. This result and the high positive correlation between days-to-heading and culm width suggest that late maturation can increase culm width and explain why the culm length of the extremely late-flowering Kazetachi is more than $2.5\times$ that of the mid-parent value.

QTLs for culm and panicle numbers

The only QTL for culm number that we detected in all 3 years was *qCN1*. The co-localization of the QTLs for culm number and days-to-heading and the phenotypic correlation between the traits imply that early-flowering plants would have more culms than late-flowering plants. The single culm of the extremely late-flowering Kazetachi supports this idea.

The numbers of culms and panicles directly contributed to dry matter yield. Twenty-five QTLs for tiller number have so far been detected in sorghum (Mace and Jordan 2011). The QTL with the largest LOD value

is *Qtinb.uga-D*, with a value of only 6.6 (Feltus et al. 2006). Their relatively low confidence values suggest that tiller number is affected by environment.

We detected two QTLs for culm number and six for panicle number, but QTLs, *qCN1* and *qPN1* with largest LOD values were co-localized with *qDHI* (days-to-heading) (Fig. 2; Table 3). The direction of the additive effect of *qDHI* and correlation analysis revealed that early-flowering lines had more culms and panicles than late-flowering lines. However, *PhyB*, a candidate gene for *qDHI*, positively regulated bud outgrowth in response to light signals (Kebrom et al. 2006, 2010); plants with a loss-of-function allele of *PhyB* had less culms (Kebrom et al. 2006). This contradiction implies that other loci nearby *qDHI* might affect the number of culm and panicle. Further analysis will reveal whether culm number is affected by pleiotropic effects of *qDHI* or by tight linkage with other loci.

Three other QTLs, *qPN4*, *qPN8.1*, and *qPN8.2* for panicle number were not associated with QTLs for days-to-heading (Table 3). As these three QTLs were detected in only 1 year, they need to be verified.

Even with successive selection for longer and wider culms, there is a limit to biomass production by only one culm, and the risk of lodging increases. The introduction of QTLs for tiller number would get around this limit. QTLs found in other *Sorghum* species with high tillering potential would be useful (Paterson et al. 1995).

Conclusion

QTL analysis using RILs derived from the parents of the F₁ cultivar Kazetachi showed that the Kazetachi phenotypes could arise from the combination of at least two major QTLs for days-to-heading and three QTLs for culm length. The co-localization of the two maturation QTLs with QTLs for other traits and the significant correlations among these traits explain their strong effects on phenotype. These maturation QTLs could prove useful in controlling biomass-related traits such as days-to-heading and culm length. Future studies will need to explain whether the co-localization is caused by tight linkage of multiple loci or by pleiotropic effects of a single QTL.

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